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Flossing apparatus for a dental bridge

Abstract

A flossing apparatus for flossing between a dental bridge and a gum of a user includes a body, an arm, and a clasp. The arm and the clasp are coupled to and extend away from the body in a same direction. The arm and the clasp are also spaced across from each other. A rod is coupled to the arm, and a distal end of the rod with respect to the rod is releasably securable to the clasp. The rod is insertable between the dental bridge and the gum to floss between the dental bridge and the gum while the distal end is secured to the clasp.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(2) Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

(3) Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

(4) Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

(5) Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

(6) The disclosure relates to flossing apparatuses and more particularly pertains to a new flossing apparatus for flossing between a dental bridge and a gum of a user.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

(7) The prior art provides myriad flossing apparatuses which include a floss segment mounted between a pair of arms such that a handle attached to the pair of arms may be positioned to slide the floss segment between teeth to remove foreign matter from between the teeth. However, such apparatuses do not provide a structure that is capable of inserting between a dental bridge and a gum of a user without removing the dental bridge. A new apparatus may address this problem by

providing a thin rod that is permanently fastened on one end and securable on another end. The rod of such an apparatus would then be able to be threaded between the dental bridge and the gum without removing the dental bridge from the user's mouth. Then the other end of the rod may be secured such that the rod is supported on both ends and may be moved along the dental bridge and the gum to remove foreign matter between the dental bridge and the gum.

BRIEF SUMMARY OF THE INVENTION

(8) An embodiment of the disclosure meets the needs presented above by generally comprising a body and an arm which is coupled to and extends away from the body. A clasp is coupled to and extends away from the body, too. The clasp and the arm extend in a same direction, and the clasp is spaced from the arm. The clasp is movable between an open configuration and a closed configuration. A rod is coupled to and extends away from the arm. The rod is positionable in a securement position when the clasp is positioned in the closed configuration. The securement position is defined by a distal end of the rod with respect to the arm is releasably secured to the clasp. The distal end of the rod is movable away from the clasp to define a released position of the rod wherein the rod and the clasp define an opening to a space between the arm and the clasp. The rod has a width such that the rod is configured to be insertable between a dental bridge and a gum of a user.

(9) There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

(10) The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

(1) The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

(2) FIG. 1 is a perspective view of a flossing apparatus according to an embodiment of the disclosure.

(3) FIG. 2 is a first side view of an embodiment of the disclosure showing a rod being positioned in a securement position.

(4) FIG. 3 is a second side view of an embodiment of the disclosure showing the rod being positioned in the securement position.

(5) FIG. 4 is a first side view of an embodiment of the disclosure showing the rod being positioned in a released position.

(6) FIG. 5 is a front view of an embodiment of the disclosure showing the rod being positioned in the securement position.

(7) FIG. 6 is a front in-use view of an embodiment of the disclosure showing the rod being positioned in the released position.

(8) FIG. 7 is an in-use view of an embodiment of the disclosure of the rod being positioned in the released position and being threaded between a dental bridge and a gum. The dental bridge is shown exploded away from the gum.

(9) FIG. 8 is an in-use view of an embodiment of the disclosure of the rod being positioned in the securement position and extending between the dental bridge and the gum. The dental bridge is

shown exploded away from the gum.

DETAILED DESCRIPTION OF THE INVENTION

(10) With reference now to the drawings, and in particular to FIGS. **1** through **6** thereof, a new flossing apparatus embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

(11) As best illustrated in FIGS. **1** through **6**, the flossing apparatus **10** generally comprises a body **12** which is elongated and defines a handle **14**. An arm **16** is coupled to and extends away from the body **12** and forms an obtuse angle with the body **12**. A clasp **18** is coupled to and extends away from the body **12**. The clasp **18** and the arm **16** extend in a same direction and are spaced from each other. The clasp **18** is movable between an open configuration **20** and a closed configuration **22** and is biased toward the closed configuration **22**.

(12) A rod **24** is coupled to and extends away from the arm **16**. The rod **24** is positionable in a securement position **26** when the clasp **18** is positioned in the closed configuration **22**. The securement position **26** is defined by a distal end **28** of the rod **24** with respect to the arm **16** being releasably secured to the clasp **18**. The distal end **28** of the rod **24** is movable away from the clasp **18** to define a released position **30** of the rod **24** wherein the rod **24** and the clasp **18** define an opening **32** to a space **34** between the arm **16** and the clasp **18**. The rod **24** has a width such that the rod **24** is configured to be insertable between a dental bridge **40** and a gum **38** of a user. The distal end **28** of the rod **24** is biased toward the clasp **18** and comprises a resiliently bendable material. The rod **24** is sufficiently bendable that the rod **24** is biased toward the securement position **26**, but not so bendable such that the rod **24** is movable away from the securement position **26** when the clasp **18** is positioned in the closed configuration **22**.

(13) In some embodiments, the rod **24** may comprise a composite of materials, wherein only a portion of the rod **24** adjacent to the arm **16** comprises the resiliently bendable material. In other embodiments, the rod **24** may comprise only a rigid material and may be pivotally coupled to the arm **16**. In such embodiments, a torsion spring or other conventional biasing member may be employed to bias said distal end **28** of said rod **24** toward said clasp **18**.

(14) An actuator **44** is coupled to the rod **24** and is movably coupled to the body **12** to move the rod **24** away from the securement position **26**. The actuator **44** is positioned between the arm **16** and the clasp **18**. The actuator **44** is also positioned adjacent to the arm **16** and is movable parallel to the arm **16** such that the actuator **44** does not interfere with the space **34** between the arm **16** and the clasp **18**. The actuator **44** is slidably received into the body **12**, and a grip **46** is coupled to the actuator **44**. The grip **46** has a pair of gripping surfaces **48** facing in opposing directions away from the body **12**. Each gripping surface **48** of the pair of gripping surfaces **48** is positioned between said arm **16** and said clasp **18**. In some embodiments, only one gripping surface **48** is provided.

(15) The clasp **18** comprises a first prong **50** and a second prong **52**, which are coupled to and extend away from the body **12**. The first prong **50** and the second prong **52** are also spaced from each other to form a groove **54** therebetween. The first prong **50** has a notch **56** extending into an outer surface **58** of the first prong **52** opposite the groove **54**. The notch **56** has a size such that the notch **56** is configured for receiving a fingernail **60**. The clasp **18** further comprises a pair of first engagement members **62** and a pair of second engagement members **64**. Each first engagement member **62** of the pair of first engagement members **62** is coupled to the first prong **50** and extends toward the second prong **52**. Each second engagement member **64** of the pair of second engagement members **64** is coupled to the second prong **52** and extends toward the first prong **50**. Each second engagement member **64** of the pair of second engagement members **64** engages an associated one of the pair of first engagement members **62** with a snap fit when the clasp **18** is positioned in the closed configuration **22**, thereby retaining the clasp **18** in the closed configuration **22**.

(16) The first prong **50** is movable away from the second prong **52** to move the clasp **18** from the closed configuration **22** to the open configuration **20** and disengage each second engagement

member **64** of the pair of second engagement members **64** from the associated first engagement member **62**. The first prong **50** may be moved, for example, by the fingernail **60** entering the notch **56** of the first prong **50** and urging the first prong **50** via the fingernail **60** away from the second prong **52**. The first prong **50** may be resiliently bendable so that the first prong **50** moves toward the second prong **52** when released to form the closed configuration **22**. The notch **56** may also cause a portion of the first prong **50** to have a thickness less than a thickness of the second prong **52** such that the first prong **50** is adapted to bend more readily than the second prong **52**.

(17) The clasp **18** further comprises a pair of first protrusions **66** and a pair of second protrusions **68**. Each first protrusion **66** of the pair of first protrusions **66** is coupled to the first prong **50** and extends toward the second prong **52**, and each second protrusion **68** of the pair of second protrusions **68** is coupled to the second prong **52** and extending toward the first prong **50**. The pair of first protrusions **66** and the pair of second protrusions **68** are positioned between the pair of first engagement members **62**. The distal end **28** of the rod **24** is received into the groove **54** and is positioned between the pair of first engagement members **62** and the pair of second engagement members **64** when rod **24** is positioned in the securement position **26**. The pair of first protrusions **66** and the pair of second protrusions **68** grip the rod **24** to prevent disengagement of the rod **24** and the clasp **18** when the rod **24** is positioned in the securement position **26**.

(18) The clasp **18** may comprise a conventional structure such as a clamp, a bolt snap, a carabiner, or a clip. In some arrangements, the rod **24** may have a snap fit directly with the clasp **18**.

(19) In use, the rod **24** is positioned in the released position **30** such that the distal end **28** of the rod **24** is free from the clasp **18**. The rod **24** is inserted between the dental bridge **40** and the gum **38** of the user while the dental bridge **40** is attached to the gum **38** via one or more dental implants **42** or existing teeth. The clasp **18** is opened to the open configuration **20**, and the distal end **28** of the rod **24** is moved into the groove **54** of the clasp **18**. The clasp **18** is closed such that it grips the rod **24** and secures the distal end **28** of the rod **24** in the groove **54** of the clasp **18**, thereby retaining the rod **24** in the securement position **26**. The flossing apparatus is moved via the handle **14** to move the rod **24** against the gum **38** and the dental bridge **40** to remove foreign matter from the gum **38** and the dental bridge **40**. The clasp **18** is opened, and the rod **24** is removed from between the dental bridge **40** and the gum **38**.

(20) With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

(21) Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. A flossing apparatus for flossing between a dental bridge and a gum of a user, the apparatus comprising: a body; an arm being coupled to and extending away from the body; a clasp being coupled to and extending away from the body, the clasp and the arm extending in a same direction,

the clasp being spaced from the arm, the clasp being movable between an open configuration and a closed configuration; and a rod being coupled to and extending away from the arm, the rod being positionable in a securement position when the clasp is positioned in the closed configuration, the securement position being defined by a distal end of the rod with respect to the arm being releasably secured to the clasp, the distal end of the rod being movable away from the clasp to define a released position of the rod wherein the rod and the clasp define an opening to a space between the arm and the clasp, the rod having a width such that the rod is configured to be insertable between the dental bridge and the gum of the user; wherein the distal end of the rod is biased toward the clasp; and further comprising an actuator being coupled to the rod and being movably coupled to the body to move the rod away from the securement position, the actuator being positioned between the arm and the clasp, the actuator being positioned adjacent to the arm and being movable parallel to the arm such that the actuator does not interfere with the space between the arm and the clasp.

2. The apparatus of claim 1, wherein the body is elongated and defines a handle.

3. The apparatus of claim 2, wherein the arm forms an obtuse angle with the body.

4. The apparatus of claim 1, wherein the clasp is biased toward the closed configuration.

5. The apparatus of claim 4, wherein the rod comprises a resiliently bendable material.

6. The apparatus of claim 5, wherein the actuator is slidably received into the body.

7. The apparatus of claim 5, further comprising a grip being coupled to the actuator, the grip having a pair of gripping surfaces facing in opposing directions away from the body.

8. The apparatus of claim 7, wherein the first prong has a notch extending into an outer surface of the first prong opposite a groove, the notch having a size such that the notch is configured for receiving a fingernail.

9. The apparatus of claim 7, wherein the clasp further comprises: a pair of first engagement members being coupled to the first prong and extending toward the second prong; a pair of second engagement members being coupled to the second prong and extending toward the first prong, each second engagement member of the pair of second engagement members engaging an associated one of the pair of first engagement members with a snap fit when the clasp is positioned in the closed configuration, the first prong being movable away from the second prong to move the clasp from the closed configuration to the open configuration and disengage each second engagement member of the pair of second engagement members from an associated first engagement member; a pair of first protrusions being coupled to the first prong and extending toward the second prong, the pair of first protrusions being positioned between the pair of first engagement members; a pair of second protrusions being coupled to the second prong and extending toward the first prong, the pair of second protrusions being positioned between the pair of second engagement members; and the distal end of the rod being positioned between the pair of first engagement members and between the pair of second engagement members when the rod is positioned in the securement position, the pair of first protrusions and the pair of second protrusions gripping the rod to prevent disengagement of the rod and the clasp when the rod is positioned in the securement position.

10. The apparatus of claim 1, wherein the clasp comprises a first prong and a second prong being coupled to and extending away from the body, the first prong and the second prong being spaced from each other to form a groove therebetween, the distal end of the rod being received into the groove when rod is positioned in the securement position.

11. A flossing apparatus for flossing between a dental bridge and a gum of a user, the apparatus comprising: a body being elongated and defining a handle; an arm being coupled to and extending away from the body, the arm forming an obtuse angle with the body; a clasp being coupled to and extending away from the body, the clasp and the arm extending in a same direction, the clasp being spaced from the arm, the clasp being movable between an open configuration and a closed configuration, the clasp being biased toward the closed configuration; a rod being coupled to and extending away from the arm, the rod being positionable in a securement position when the clasp is

positioned in the closed configuration, the securement position being defined by a distal end of the rod with respect to the arm being releasably secured to the clasp, the distal end of the rod being movable away from the clasp to define a released position of the rod wherein the rod and the clasp define an opening to a space between the arm and the clasp, the rod having a width such that the rod is configured to be insertable between the dental bridge and the gum of the user, the distal end of the rod being biased toward the clasp, the rod comprising a resiliently bendable material; an actuator being coupled to the rod and being movably coupled to the body to move the rod away from the securement position, the actuator being positioned between the arm and the clasp, the actuator being positioned adjacent to the arm and being movable parallel to the arm such that the actuator does not interfere with the space between the arm and the clasp, the actuator being slidably received into the body; a grip being coupled to the actuator, the grip having a pair of gripping surfaces facing in opposing directions away from the body; and wherein the clasp comprises: a first prong and a second prong being coupled to and extending away from the body, the first prong and the second prong being spaced from each other to form a groove therebetween, the first prong having a notch extending into an outer surface of the first prong opposite the groove, the notch having a size such that the notch is configured for receiving a fingernail; a pair of first engagement members being coupled to the first prong and extending toward the second prong; a pair of second engagement members being coupled to the second prong and extending toward the first prong, each second engagement member of the pair of second engagement members engaging an associated one of the pair of first engagement members with a snap fit when the clasp is positioned in the closed configuration, the first prong being movable away from the second prong to move the clasp from the closed configuration to the open configuration and disengage each second engagement member of the pair of second engagement members from an associated first engagement member; a pair of first protrusions being coupled to the first prong and extending toward the second prong, the pair of first protrusions being positioned between the pair of first engagement members; a pair of second protrusions being coupled to the second prong and extending toward the first prong, the pair of second protrusions being positioned between the pair of second engagement members; and the distal end of the rod being received into the groove when rod is positioned in the securement position, the distal end of the rod being positioned between the pair of first engagement members and between the pair of second engagement members when the rod is positioned in the securement position, the pair of first protrusions and the pair of second protrusions gripping the rod to prevent disengagement of the rod and the clasp when the rod is positioned in the securement position.
